Should Platforms Be Allowed to Sell on Their Own Marketplaces?

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Motivation: A Dual Role platform

Many large platforms now play two roles at the same time.
(Eg. Amazon, JD.com, Apple App Store...)

- They run a marketplace for third-party sellers.
- They also sell their own products on the same marketplace.

The key tension

The platform is both the market **organizer** and one of the **sellers** in the market.

Why Regulators Worry

The concern is not only that the platform sells its own products.

- **Product imitation:** the platform may use marketplace data to **copy** successful sellers.
- **Self-preferring:** the platform may **steer** consumers toward its own products.
- **Innovation harm:** third-party sellers may invest less if they expect imitation or steering.

Policy question

Should regulators ban the platform from selling on its own marketplace?

Research Question

The paper asks three linked questions.

1. Does **banning** dual mode always **raise consumer surplus and welfare**?
2. If dual mode is banned, does the platform **switch to seller mode or marketplace mode**?
3. Are targeted **rules on imitation and self-preferring** better than a full ban?

Key Insight

The problem is not dual mode itself, but how the platform uses its power.

- A ban on dual mode can hurt consumers and welfare.
- Targeting bad conduct is often better.

Roadmap for the Talk

1. Motivation and main question
2. Baseline model setup
3. Baseline equilibria and price squeeze
4. Ban on dual mode
5. Imitation, self-preferring, and policy comparison

How to read the model

Each mathematical result is followed by its economic intuition.

Players

The model studies one product category.

- Platform M: can run a marketplace, sell its own product, or do both.
- Innovative seller S: has a better product and can invest in quality.
- Fringe sellers: sell a basic product and compete strongly.
- Consumers: each buys at most one unit or takes an outside option.

Intuition

The platform is important because consumers use it to find products and to trade more easily.

Values and Parameters

Product values are:

Product	Consumer value	Meaning
Fringe sellers	v	Basic product
Innovative seller S	$v + \Delta$	Better product
Platform M	$v + \sigma$	Platform's own product

- $b > 0$: convenience benefit from buying through the platform.
- $v_o \sim G$: outside option.
- τ: platform commission.

Intuition

$\sigma > 0$ means the platform is a better seller than fringe sellers; $\sigma < 0$ means the opposite.

Innovation by Seller S

Seller S chooses innovation $\Delta \geq \Delta^l$ and pays fixed cost $K(\Delta)$.

$$K'(\bar{\Delta}) = G(\nu + \sigma + b)$$

Meaning of $\bar{\Delta}$

$\bar{\Delta}$ is the high innovation level that can arise when the market gives strong rewards to innovation.

Intuition

The left side is marginal cost of innovation. The right side is the demand gain from better quality.

Product Discovery

Consumers do not know about S at the start.

- Consumers know the fringe sellers.
- Consumers may know the platform's own product.
- Consumers discover S only if S is available on the platform.

Why this matters

The platform is not only a sales channel. It is also a discovery channel.

Intuition

If a policy changes the platform's mode, it may also change whether consumers find S.

Three Platform Modes

Mode	What M does	Key feature
Marketplace	Hosts third-party sellers	Earns commission τ
Seller	Sells its own product	No marketplace for S
Dual	Hosts sellers and sells own product	Both roles at once

Intuition

The policy debate is about whether the third mode should be allowed.

Timing

The game is solved by backward induction.

1. M chooses its mode and sets commission τ if there is a marketplace.
2. S decides whether to join the platform and chooses Δ.
3. Sellers set prices.
4. Consumers choose what to buy.

Equilibrium concept

Subgame Perfect Nash Equilibrium.

Marketplace Mode

In marketplace mode, M only charges commission τ.

$$p_i \leq \tau + \Delta, \qquad p_o \leq \tau + \Delta - b$$

Showrooming constraint

If $\tau > b$, seller S can use the platform for discovery and then sell directly.

Intuition

The platform can charge at most the extra convenience value b that it gives consumers.

Marketplace Equilibrium

Proposition 1:

$$\tau^{mkt} = b, \qquad G(\nu) = K'(\Delta^{mkt}), \qquad p_i^* = b + \Delta^{mkt}$$

$$\Pi^{mkt} = bG(\nu), \qquad \pi^{mkt} = \Delta^{mkt}G(\nu) - K(\Delta^{mkt})$$

Intuition

M sets the highest fee that does not push trade outside the platform. S invests until marginal cost equals marginal market gain.

Seller Mode

In seller mode, the platform only sells its own product.

$$\max_{p_m \leq b+\sigma} p_m G(v + b + \sigma - p_m)$$

The competitive constraint binds:

$$p_m^* = b + \sigma$$

Proposition 2

M sells to all consumers, while S sells to no one.

Intuition

The platform captures its convenience value and its product advantage, but consumers do not discover S.

Dual Mode: Two Pricing Cases

In dual mode, M both charges commission and sells its own product.

Case 1: $\sigma \geq \Delta$

The platform's own product is good enough. Consumers buy from M.

$$p_i^* = \tau, \qquad p_m^* = \tau + \sigma - \Delta$$

Case 2: $\Delta > \sigma$

Seller S has the better product. But M still puts pressure on S's price.

Intuition

Dual mode matters most when S is better, but M can still discipline S's price.

Price Squeeze

In the price squeeze equilibrium:

$$p_i^* = p_m^* + \Delta - \sigma$$

$$p_m^* \in [\max\{\tau - \Delta + \sigma, 0\}, \tau + \min\{\sigma, 0\}]$$

$$\Pi = \tau G(v + \sigma + b - p_m^*)$$

Intuition

M may not sell, but its own product caps S's price. A lower S price raises demand and commission income.

Innovation Constraint

Define $\bar{\tau}$ by:

$$(\bar{\Delta} - \sigma - \bar{\tau})G(\nu + b + \sigma) - K(\bar{\Delta}) = 0$$

Lemma 1:

- If $\tau \leq \bar{\tau}$, then S chooses high innovation: $\Delta = \bar{\Delta}$.
- If $\tau > \bar{\tau}$, then S chooses low innovation: $\Delta = \Delta^l$.

Intuition

A high commission lowers S's margin. If the margin is too low, innovation is not worth the cost.

Dual Mode Equilibrium

Proposition 3: M balances two limits.

- Showrooming constraint: $\tau \le b$.
- Innovation constraint: $\tau \le \bar{\tau}$.

If high innovation is profitable enough:

$$\tau^{dual} = \min\{b, \bar{\tau}\}, \qquad \Delta^{dual} = \bar{\Delta}$$

Otherwise:

$$\tau^{dual} = b, \qquad \Delta^{dual} = \Delta^{l}$$

Intuition

The platform may charge less to keep S willing to innovate, because innovation expands the market.

Which Mode Does the Platform Choose?

Corollary 1:

- M prefers marketplace mode over seller mode iff $\sigma \leq 0$.
- Dual mode is always better for M than marketplace mode.
- There is a threshold $\underline{\sigma} > 0$ such that:

$$\Pi^{dual} > \Pi^{sell} \quad \text{iff} \quad \sigma < \underline{\sigma}$$

Intuition

Dual mode creates more transactions through price squeeze. But if M's own product is much better, M may prefer to sell alone.

What Does a Ban Change?

If dual mode is banned, M must choose one pure mode.

- If $\sigma > 0$, seller mode is more attractive.
- If $\sigma \leq 0$, marketplace mode is more attractive.

Important point

The platform reacts to the policy. It does not keep the same behavior.

Ban in the Baseline Model

Proposition 4:

Region	New mode	Consumer surplus	Welfare
$\sigma \geq \underline{\sigma}$	Seller	no change	no change
$0 < \sigma < \underline{\sigma}$	Seller	lower	lower
$\sigma \leq 0$	Marketplace	lower	unclear

Intuition

When the ban matters, consumers lose the price squeeze effect. If M switches to seller mode, consumers may also lose discovery of S.

Why Welfare Can Fall

The key case is $0 < \sigma < \underline{\sigma}$.

- Before the ban, M uses dual mode.
- After the ban, M switches to seller mode.
- Consumers no longer discover S through the platform.
- Price competition from S is weaker.
- Consumers cannot combine S's better product with platform convenience b.

Intuition

A rule that looks like it protects sellers can hurt consumers, because it may close the discovery channel.

Adding Two Bad Conducts

The paper then adds the two main policy concerns.

Product imitation

If S joins the platform, M can copy S's product before prices are set.

Self-preferring

After prices are set, M chooses whether to show S to consumers.

Intuition

These assumptions give the strongest case against dual mode: copying is perfect and steering is perfect.

Dual Mode with Imitation and Steering

Proposition 5:

$$\Delta^{dual} = \Delta^{l}$$

If $\sigma \geq \Delta^{l}$:

$$\Pi^{dual} = (b + \sigma)G(\nu)$$

If $\sigma < \Delta^{l}$:

$$\Pi^{dual} = (b + \Delta^{l})G(\nu)$$

Intuition

S does not invest because high innovation can be copied or hidden. M can extract the value through high fees or imitation.

Does a Dual Mode Ban Fix This?

Proposition 6:

- If $\sigma \geq \Delta^l$: no effect.
- If $0 < \sigma < \Delta^l$: M switches to seller mode; welfare falls.
- If $\sigma \leq 0$: M switches to marketplace mode; innovation and welfare rise.

Intuition

A structural ban does not directly restore innovation or fair display. It only forces M to pick one pure mode.

Policy 1: Ban Self-Preferencing

If self-preferring is banned, M must show S when S is listed.

- On-platform competition returns.
- The showrooming constraint returns.
- Final prices fall.
- But imitation is still possible, so innovation may stay low.

Intuition

This policy fixes the display problem, but not the copying problem.

Policy 2: Ban Imitation

If imitation is banned, M can commit not to copy S.

$$\tau \leq \bar{\tau}$$

If innovation is valuable enough:

$$\tau = \bar{\tau}, \qquad \Delta = \bar{\Delta}$$

Intuition

Commitment not to copy can make S invest more. A larger market can also help the platform.

Policy 3: Ban Both

If both imitation and self-preferring are banned, the model is close to the baseline dual mode.

- S is shown to consumers.
- M cannot copy S.
- Price competition returns.
- Innovation incentives return.

Intuition

The two rules work together: one restores competition, the other restores innovation.

Policy Comparison

Main policy lesson:

Policy	What it targets	Main effect
Ban dual mode	Market structure	Can remove useful dual mode benefits
Ban self-preferring	Display choice	Restores price competition
Ban imitation	Copying choice	Restores innovation incentives
Ban both	Both conducts	Best at keeping benefits and limiting harm

Intuition

The problem is not always dual mode itself. The problem is how the platform uses its power.

Main Takeaways

1. Dual mode can create a price squeeze that lowers prices and raises demand.
2. Banning dual mode can lower consumer surplus and welfare.
3. Imitation hurts innovation incentives.
4. Self-preferring weakens platform competition.
5. Targeted conduct rules are often better than a full structural ban.

Final message

Regulate the harmful conduct, not only the platform's dual role.